PROJECT–SDG ALIGNMENT

Programme Outcomes (POs) • Programme Specific Outcomes (PSOs) • Sustainable Development Goals (SDGs)

Student and Subject Details

Student Name	**GUBBALA CHAITANYA SAI KRISHNA**
Roll Number	**24761A0517**
Year / Semester / Section	IV Year / I Semester / B.Tech CSE
Department / Programme	Department of Computer Science and Engineering / B.Tech
Subject Name and Code	Full Stack Web Development & Project Work (CS701)
Faculty Name	Faculty In-Charge / Project Guide (Department of CSE)
Project Title / Assignment / Case Study	**Vehicle Rental System (RentRide) — Full-Stack Web Application**
Technology / Tools Used	Java 21 LTS, Java EE Servlets, JSP, JDBC, MySQL 8.0, Apache Tomcat 9.0, jBCrypt 0.4, Vanilla CSS3, Selenium WebDriver 4.x

Project / Subject Activity Summary

1. Problem statement / need addressed:

Traditional vehicle rental operations suffer from manual record-keeping inaccuracies, lack of real-time inventory visibility, double-booking collisions, and opaque pricing structures. Offline rental processes create customer friction, manual billing discrepancies, and insecure credential storage. The RentRide Vehicle Rental System addresses these challenges by delivering an automated, web-based digital ecosystem providing transparent self-service vehicle discovery, concurrency-protected reservations, automated price estimation, and centralized fleet governance.

2. Main objectives:

• Objective 1: Develop an automated, intuitive self-service portal for browsing, inspecting, and reserving multi-category fleet units.
• Objective 2: Enforce strict transactional database concurrency with row-level pessimistic locking (SELECT ... FOR UPDATE) to eliminate double-booking overlaps.
• Objective 3: Implement secure authentication via adaptive BCrypt salted password hashing and role-based access control (RBAC).
• Objective 4: Build a centralized operational dashboard for real-time KPI metrics, gross revenue calculation, and fleet governance.
• Objective 5: Create an interactive mock payment gateway sandbox simulating Credit/Debit Card, UPI, and Netbanking transactions.

3. Brief description of the solution and its key features:

RentRide is an enterprise-grade full-stack web application engineered on a classic Three-Tier Architectural model (JSP presentation, Java EE Servlet controllers, and MySQL 8.0 persistence via JDBC). Key features include: real vehicle photography cards

with transparent daily rates; dynamic client-side rental price computation; atomic reservation booking with automatic date overlap rejection via pessimistic row locking; a mock development payment sandbox (Card, UPI, Netbanking); customer reservation dashboards with self-service cancellation; and an executive administration console with gross revenue KPI analytics and fleet governance controls.

4. Intended users / beneficiaries:

• Individual Customers & Urban Commuters: Commuters, business professionals, and travelers seeking affordable, transparent, on-demand vehicle rentals with instantaneous booking confirmations and personal reservation dashboards.
• Fleet Managers & Rental Business Operators: Commercial rental agencies and fleet administrators who gain real-time visibility into vehicle utilization, gross revenue analytics, automated reservation approval, and centralized fleet maintenance.

5. Expected or observed outcomes / evidence:

The application was deployed on Apache Tomcat 9.0 (port 8084) backed by MySQL 8.0 (port 3306) and empirically validated through 15 automated end-to-end functional test cases with a 100.0% pass rate. Observed outcomes demonstrate: zero duplicate reservations under concurrent overlapping date submissions; sub-3-second server boot time; instantaneous price calculation across arbitrary date windows; seamless authentication with zero plaintext password storage via BCrypt salted cryptography; and verified multi-channel simulated payment settlement.

Mapping with Programme Outcomes (POs)

Instructions: Select only the POs meaningfully demonstrated by the work. Describe the specific activity, design decision, analysis, or evidence supporting each mapping.

PO No.	Relevant PO	How the work demonstrates this PO
PO1	Engineering knowledge	Applied mathematical models for calendar duration and rental pricing calculation (Days * Daily Rate). Leveraged core computer science and software engineering principles in Java 21, object-oriented design patterns (MVC, DAO), and relational database normalization (3NF).
PO2	Problem analysis	Formulated solutions for the critical double-booking concurrency challenge in scheduling systems. Analyzed date-interval intersection logic (start_date < ? AND end_date > ?) and implemented pessimistic row locking (SELECT ... FOR UPDATE) to eliminate race conditions.
PO3	Design/development of solutions	Architected and implemented a complete Three-Tier enterprise solution meeting customer booking and administrator governance requirements. Designed modular servlet controllers, JDBC DAOs, and responsive JSP presentation views.
PO4	Conduct investigations of complex problems	Conducted systematic investigations into date overlap collisions and transaction rollbacks. Verified application correctness across 15 comprehensive automated test cases, empirically validating data integrity and session security.
PO5	Engineering Tool Usage	Employed modern software engineering tools including Java 21 LTS OpenJDK, Apache Tomcat 9.0.120, MySQL Community Server 8.0, Selenium WebDriver 4.x with headless Microsoft Edge, and Python test harnesses.

PO6	The Engineer and the World	Addressed societal transportation challenges by facilitating the sharing economy, optimizing vehicle utilization to reduce urban traffic congestion, reliance on private car ownership, and environmental pollution.
PO7	Ethics	Enforced professional ethics by protecting user privacy, implementing salted BCrypt password hashing without plaintext storage, preventing privilege escalation via RBAC, and ensuring fully transparent pricing without hidden charges.
PO8	Individual and Collaborative Team Work	Engineered modular software components partitioned into distinct packages (model, dao, controller, util, webapp), establishing clean interfaces and separation of concerns that support multi-developer collaborative engineering.
PO9	Communication	Generated comprehensive technical documentation, detailed relational schemas, empirical test results, and responsive UI feedback with explicit error messages and validation alerts.
PO10	Project management and finance	Incorporated financial tracking mechanisms including dynamic rate calculation, multi-method simulated payment processing, and administrator gross revenue aggregation for commercial viability.
PO11	Life-long learning	Demonstrated self-directed learning and adaptation by mastering modern Java EE servlet specifications, JDBC concurrency locking, modern CSS custom properties, and automated headless browser verification techniques.

Mapping with Programme Specific Outcomes (PSOs)

Instructions: Select the PSOs demonstrated by the work and explain the connection to the approved programme-specific outcome.

PSO No.	How the work demonstrates this PSO
PSO1: The ability to apply Software Engineering practices and strategies in software project development using open-source programming environment for the success of organization.	Applied standard software engineering practices (MVC architecture, DAO design pattern, thread-safe connection management, unit testing) entirely within an open-source development stack (Java OpenJDK, Apache Tomcat, MySQL Community Edition).
PSO2: The ability to design and develop computer programs in networking, web applications and IoT as per the society needs.	Engineered a robust multi-tier web application operating over standard HTTP/1.1 networking protocols, managing session states via secure HTTP cookies, and addressing societal needs for accessible, affordable urban transportation.

PSO3: To inculcate an ability to analyze, design and implement database applications.	Analyzed, designed, and implemented a normalized relational MySQL database schema (users, vehicles, bookings, payments) with foreign key constraints, cascading referential actions, indexed queries, and row-level pessimistic locking via JDBC.

Mapping with Sustainable Development Goals (SDGs)

Instructions: Review the 17 SDGs below. Select the one or more goals that are directly supported by the project. For each selected goal, explain the specific feature or outcome and provide evidence where possible. Do not claim an SDG connection based only on the project title. (Select at Max: 3 to 4 SDGs)

Select (✓)	SDG No.	Official goal title	Reason for selection / project contribution
☐	SDG 1	No Poverty	–
☐	SDG 2	Zero Hunger	–
☐	SDG 3	Good Health and Well-being	–
☐	SDG 4	Quality Education	–
☐	SDG 5	Gender Equality	–

☐	SDG 6	Clean Water and Sanitation	–
☐	SDG 7	Affordable and Clean Energy	–
☑	SDG 8	Decent Work and Economic Growth	Enables local car rental agencies and fleet owners to digitize their business operations, improve operational efficiency, automate bookkeeping, and expand their economic reach through digital commerce channels.
☑	SDG 9	Industry, Innovation and Infrastructure	Modernizes traditional transportation services through innovative digital infrastructure, open-source enterprise architecture, and real-time database-driven reservation technology.
☐	SDG 10	Reduced Inequalities	–
☑	SDG 11	Sustainable Cities and Communities	Promotes sustainable urban shared mobility by optimizing the utility and occupancy of shared vehicle fleets, thereby reducing the demand for personal vehicle ownership, minimizing traffic congestion, and curbing vehicular exhaust emissions. (Primary SDG)
☑	SDG 12	Responsible Consumption and Production	Fosters the circular and sharing economy by maximizing the operational lifecycle of vehicles through shared access rather than manufacturing excess private vehicles, minimizing resource consumption and automotive waste.
☐	SDG 13	Climate Action	–
☐	SDG 14	Life Below Water	–
☐	SDG 15	Life on Land	–
☐	SDG 16	Peace, Justice and Strong Institutions	–
☐	SDG 17	Partnerships for the Goals	–

Primary SDG: SDG 11 (Sustainable Cities and Communities) Secondary SDG(s): SDG 8, SDG 9, SDG 12

Specific project feature(s) linked to the selected SDG(s):

RentRide directly fosters sustainable urban mobility (SDG 11) by offering on-demand shared vehicular access across multiple vehicle classifications (Hatchback, Sedan, SUV, Motorcycle). Centralized fleet inventory and real-time scheduling minimize idle vehicle downtime and maximize asset utilization (SDG 12). The digital booking pipeline and payment sandbox eliminate physical paper contracts and manual bookkeeping (SDG 9), while empowering local rental operators with automated financial tracking and business scalability (SDG 8).

Measurable or verifiable contribution (if available):

Empirically verified across 15 automated test cases, the system optimizes fleet utilization by preventing duplicate booking downtime through pessimistic row locking. Shared vehicle access directly reduces the demand for individual vehicle purchases,

curbing urban traffic congestion and vehicular emissions. Furthermore, the 100% digital paperless web workflow eliminates paper rental slips, contracts, and manual receipts, achieving an energy-efficient operational model.

Student Declaration

I, GUBBALA CHAITANYA SAI KRISHNA, bearing Roll No. 24761A0517, declare that the information provided in this template accurately describes my work and its relationship to the selected Programme Outcomes, Programme Specific Outcomes, and Sustainable Development Goals. The SDG mappings are based on the project's actual objectives, features, and expected or demonstrated outcomes.

Student Signature(s): G. Chaitanya Sai Krishna

Date: